

VNUHCM – UNIVERSITY OF SCIENCE
TAKE-HOME MID-TERM RETAKE EXAMINATION

Total: 10 points

Term 3 – Academic year 2025–2026

Course: Applied Statistics for Engineers and Scientists II

Class: 24A02

Released: 31-07-2026

Due: 01-08-2026

Instructions. Use natural logarithms and significance level $\alpha = 0.05$ unless stated otherwise.

Student name: _____ **Student ID:** _____

Problem 01. Multiple linear regression: estimation, inference, and prediction (4 points)

Ten observations are modeled by

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

The sample means are

$$\bar{x}_1 = 4, \quad \bar{x}_2 = 3, \quad \bar{y} = 10.$$

Let

$$z_i = \begin{pmatrix} x_{i1} - 4 \\ x_{i2} - 3 \end{pmatrix}, \quad w_i = y_i - 10.$$

The centered cross-products are

$$G := \sum_{i=1}^{10} z_i z_i^\top = \begin{pmatrix} 20 & 10 \\ 10 & 20 \end{pmatrix}, \quad g := \sum_{i=1}^{10} z_i w_i = \begin{pmatrix} 20 \\ -5 \end{pmatrix},$$

and

$$\text{SST} := \sum_{i=1}^{10} (y_i - \bar{y})^2 = 40.$$

The reduced model containing only the intercept and x_1 has residual sum of squares

$$\text{RSS}_{\text{red}} = 20.$$

(a) Starting from the least-squares criterion, derive the centered normal equations

$$G \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = g,$$

and compute $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$. **(0.8 points)**

(b) Compute SSR, SSE, R^2 , adjusted R^2 , and the unbiased estimate $\hat{\sigma}^2$. **(0.7 points)**

(c) Define the centered intercept

$$\alpha = \beta_0 + 4\beta_1 + 3\beta_2.$$

Show that, in the parameterization $\theta = (\alpha, \beta_1, \beta_2)^\top$,

$$\widehat{\text{Var}}(\hat{\theta}) = \hat{\sigma}^2 \left(\begin{array}{c|cc} 1/10 & 0 & 0 \\ \hline 0 & & \\ 0 & & G^{-1} \end{array} \right),$$

i.e. a block-diagonal matrix with top-left entry $1/10$ and lower-right 2×2 block G^{-1} . Then transform this matrix to obtain the estimated covariance matrix of $(\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2)^\top$. **(1.0 points)**

(d) Test $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 \neq 0$ in two ways:

(i) use the coefficient t statistic;

(ii) use the nested-model F statistic based on RSS_{red} .

Verify numerically that $F = t^2$ for this one-restriction test, and state the conclusion. **(0.7 points)**

- (e) For $x_0 = (1, 5, 2)^\top$, compute the fitted response, a 95% confidence interval for the conditional mean, and a 95% prediction interval for a new response. In centered coordinates $z_0 = (1, -1)^\top$, use

$$h_0 = \frac{1}{10} + z_0^\top G^{-1} z_0,$$

$$\hat{y}_0 \pm t_{0.975,7} \hat{\sigma} \sqrt{h_0}, \quad \hat{y}_0 \pm t_{0.975,7} \hat{\sigma} \sqrt{1 + h_0},$$

and $t_{0.975,7} = 2.365$. **(0.8 points)**

Problem 02. Transformation, interpretation, and back-transformation

(3 points)

A positive response is modeled by

$$Y = \alpha \exp(\beta X) \exp(\varepsilon), \quad \varepsilon \sim N(0, \sigma^2), \quad \alpha > 0.$$

Let $V = \log Y$. For six observations, the transformed-data summaries are

$$\bar{x} = \frac{5}{2}, \quad \bar{v} = \frac{9}{4}, \quad S_{xx} = \sum_{i=1}^6 (x_i - \bar{x})^2 = \frac{35}{2},$$

$$S_{xv} = \sum_{i=1}^6 (x_i - \bar{x})(v_i - \bar{v}) = \frac{35}{4}, \quad \text{SSE}_{\log} = 0.10.$$

- (a) Transform the model into a linear regression, and compute $\hat{\alpha}$ and $\hat{\beta}$. Interpret $\exp(\hat{\beta})$ in the original response scale. **(0.75 points)**
- (b) Compute $\hat{\sigma}^2$ on the log scale and a 95% confidence interval for β . Use

$$\text{se}(\hat{\beta}) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, \quad t_{0.975,4} = 2.776.$$

Translate the interval into an interval for the multiplicative effect $\exp(\beta)$. **(0.75 points)**

- (c) At $x_0 = 6$, compute both

$$\widehat{\text{Med}}(Y \mid X = x_0) = \exp(\hat{\alpha} + \hat{\beta}x_0)$$

and the lognormal mean-corrected estimate

$$\widehat{\mathbb{E}}(Y \mid X = x_0) = \exp\left(\hat{\alpha} + \hat{\beta}x_0 + \frac{\hat{\sigma}^2}{2}\right), \quad \hat{\alpha} = \log \hat{\alpha}.$$

Explain why the two quantities differ. **(0.75 points)**

- (d) Construct a 95% prediction interval for a future response at $x_0 = 6$ by first forming the interval for $V_0 = \log Y_0$ and then exponentiating both endpoints. Use

$$h_0 = \frac{1}{6} + \frac{(x_0 - \bar{x})^2}{S_{xx}}, \quad \hat{v}_0 \pm t_{0.975,4} \hat{\sigma} \sqrt{1 + h_0}.$$

(0.75 points)

Problem 03. Logistic regression and threshold-sensitive evaluation

(2 points)

A fitted model for a binary equipment alarm is

$$\log \frac{\hat{p}(x)}{1 - \hat{p}(x)} = -5 + 0.07x_1 + 1.10x_2 - 1.30x_3,$$

where x_1 is temperature, x_2 is vibration, and $x_3 = 1$ indicates preventive maintenance.

(a) Interpret the three slope coefficients in terms of conditional odds ratios. **(0.4 points)**

(b) Compute the fitted probabilities for

$$A = (50, 1, 1), \quad B = (65, 2, 0),$$

and classify them using threshold 0.5. **(0.4 points)**

(c) On an independent test set, the observed labels and fitted probabilities are

case	1	2	3	4	5	6	7	8
Y_i	1	0	1	0	1	0	1	0
$\hat{p}_i$	0.82	0.68	0.62	0.52	0.41	0.33	0.28	0.12

For thresholds $c = 0.50$ and $c = 0.30$, classify by $\hat{Y}_i = \mathbf{1}\{\hat{p}_i \geq c\}$. For each threshold compute TP, FP, TN, FN , accuracy, precision, recall, specificity, and F_1 . **(0.8 points)**

(d) Suppose one false negative has cost 4 and one false positive has cost 1. Using empirical cost

$$L(c) = 4FN(c) + FP(c),$$

select between the two thresholds and justify the choice. **(0.4 points)**

Problem 04. A theoretical proof in multiple linear regression **(1 point)**

Let

$$Y = X\beta + \varepsilon,$$

where $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$ is deterministic with $\text{rank}(X) = p$, $\beta \in \mathbb{R}^p$, $\mathbb{E}(\varepsilon) = 0$, and $\text{Var}(\varepsilon) = \sigma^2 I_n$. Define

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

Let $\tilde{\beta} = CY$ be any linear estimator that is unbiased for β for every $\beta \in \mathbb{R}^p$, where $C \in \mathbb{R}^{p \times n}$ is deterministic. No Gaussian assumption may be used.

Prove the Gauss–Markov conclusion

$$\text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta}) \succeq 0.$$

A complete proof must:

- (i) derive $\hat{\beta} - \beta = (X^\top X)^{-1} X^\top \varepsilon$ and $\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$;
- (ii) derive $CX = I_p$ from unbiasedness for every β ;
- (iii) define $D = C - (X^\top X)^{-1} X^\top$, prove $DX = 0$, show that the cross terms vanish, and identify the positive-semidefinite covariance difference explicitly.

===== END OF EXAMINATION =====